

Petrochemical Sensor Data — Analysis Summary

Pipeline: SQL (Bronze → Silver → Gold, star schema) → Gold layer imported and merged into a single fact table → Python (EDA + statistical analysis)

1. Exploratory Data Analysis (Univariate Distributions)

Variable	Observed Pattern
Product Yield	Tree/bell-shaped distribution
Energy Intensity	Similar shape to yield, but right-skewed (more mass on the higher side)
Reactor Temperature	Similar bell-shaped histogram
Reactor Pressure	Non-uniform — low frequency (~150) below 0.7 bar, sharp rise to ~500 in the 0.7–0.8 bar band, then declining to ~350
Feedstock Flow	Bell-shaped, similar to yield
Vibration Level	Fluctuates around a frequency of ~325, no strong skew

2. Production Performance Breakdown

2.1 Unit-wise

Unit	Mean Yield	Std Dev
Ethylene_Plant_01	80.380	14.100
Ammonia_Unit_02	80.257	14.161
Methanol_Complex_03	80.005	13.978

2.2 Catalyst-wise

Catalyst	Mean Yield	Std Dev
Iron_Standard_V5	80.273495	14.230309

	Catalyst	Mean Yield	Std Dev
	Cobalt_Premium_Z2	80.273129	13.746840
	Platinum_Base_X1	80.103755	14.263458

Iron and Cobalt have virtually identical means (difference ≈ 0.0004 tons — statistically negligible); Cobalt has the lower std, i.e., more consistent yield.

2.3 Shift-wise

	Shift	Mean Yield	Std Dev
	Morning	80.599896	14.138588
	Night	80.042066	13.944690
	Evening	80.006573	14.155533

Morning shift shows the highest mean **and** lowest-ish variability — best overall performer among shifts.

2.4 Unit × Catalyst Combinations

Best (highest mean): Ethylene_Plant_01 + Iron_Standard_V5 → mean 80.5055, std 14.289

Most consistent (lowest std): Methanol_Complex_03 + Cobalt_Premium_Z2 → mean 80.2904, std 13.6640

Highest variability (least reliable): Ammonia_Unit_02 + Platinum_Base_X1 → mean 80.3116, std 14.5302

Lowest mean: Methanol_Complex_03 + Platinum_Base_X1 → mean 79.7472, std 13.9651

Selection logic: "best" depends on whether the priority is high mean yield or low variability (consistency) — these are two different, sometimes competing, criteria.

2.5 Time-wise Production

Monthly yield is stable around 80 tons — no long-term upward/downward trend.

Highest monthly average: **December 2022 (83.01 tons)**

Lowest monthly average: **February 2020 (78.18 tons)**

Monthly std ranges from 12.36 to 15.64 tons — fluctuation, not a sustained trend.

2.6 Unit × Shift

Morning shift is highest across all three units; Night is generally lowest.

Most noticeable shift variation is in Methanol_Complex_03 (Evening < Morning and Night).

Overall differences remain small — shift performance is broadly similar across units.

2.7 Unit × Catalyst × Shift

Highest: Methanol_Complex_03 + Platinum_Base_X1 + Morning → 81.62 tons

Lowest: Ammonia_Unit_02 + Cobalt_Premium_Z2 + Night → 78.69 tons

Morning-shift combinations dominate the top-performing list, reinforcing the shift-level finding.

Best-vs-worst gap across all combinations is small (~2.93 tons).

3. Catalyst Aging Effects

Relationship	Correlation (r)
Catalyst age vs. Yield	0.008 (no relation)
Catalyst age vs. Yield, by catalyst type	Cobalt: -0.003, Iron: 0.019, Platinum: 0.008 (all ~0, no relation)
Catalyst age vs. Energy Intensity	0.031 (no relation)

Conclusion: Catalyst aging shows no meaningful effect on yield or energy intensity in this dataset.

4. Operating Conditions vs. Yield (Correlations)

Variable	Relationship with Yield
Reactor Pressure	Strong positive (r = 0.779)
Feedstock Flow	Moderately strong positive (r = 0.625)
Vibration Level	Moderate negative (r = -0.381)
Reactor Temperature	No linear relationship (dense circular scatter)
Valve Opening	No linear relationship
Sensor Health Index	No relationship with yield or energy intensity
Energy Intensity	Strong negative (r = -0.819) — higher yield tends to come with better energy efficiency
Individual utility variables (standalone)	Almost no linear relationship with yield individually

At the unit level: **Ethylene_Plant_01** has the highest average yield *and* the lowest average energy intensity, though differences across units are small.

Vibration increasing → energy intensity tends to increase as well.

Sensor health shows no relation to either yield or energy intensity.

5. Deep Dive: Reactor Pressure vs. Yield

Pressure Range (bar) Count Mean Yield Std

0.560–0.648	875	59.98	7.14
0.648–0.736	1,808	69.79	7.80
0.736–0.824	2,892	76.98	8.69
0.824–0.912	2,178	86.21	9.63
0.912–1.000	2,246	94.85	10.82

Reactor pressure is the **strongest single observed factor** ($r = 0.779$); average yield rises consistently from ~60 tons to ~95 tons across the pressure bands.

But: substantial yield variation exists *within* the same pressure band — pressure alone doesn't fully determine yield.

Within the same pressure range, **higher feedstock flow** consistently distinguishes high-yield from low-yield observations. Temperature stays stable; vibration shows little difference within a given pressure band.

Takeaway: Vibration relates more to the broader pressure–yield relationship, not to within-pressure variation. Feedstock flow is the key secondary explanatory factor.

6. Deep Dive: Feedstock Flow vs. Yield

Correlation: **$r = 0.625$** (positive)

Mean yield rises steadily across flow bins: 58.94 → 69.54 → 79.96 → 90.74 → 103.13 tons

Relationship is approximately linear; no plateau observed up to ~759 m³/h.

Within the same flow range, higher-yield observations show:

Higher reactor pressure (~0.90 bar vs. ~0.72 bar for low-yield)

Lower vibration (~3.5–4.0 mm/s vs. ~4.9–5.5 mm/s)

Temperature, sensor health, and valve opening show little difference between high/low yield groups at the same flow level.

Takeaway: Flow is an important positive driver, but reactor pressure is the major *additional* factor explaining yield differences even at similar flow levels.

7. Deep Dive: Vibration vs. Yield (Equipment Health)

Relationship is **nonlinear**: yield stays relatively stable (~84.1-84.5 tons) at low-to-moderate vibration, then drops sharply to **67.6 tons** at the highest range (6.87-8.50 mm/s).

At high vibration: energy intensity rises (2.72 → 3.39), reactor pressure falls (0.85 → 0.68 bar).

Correlations: vibration vs. yield $r = -0.381$; vibration vs. energy intensity $r = 0.337$; vibration vs. flow/sensor health $\approx$ negligible.

Takeaway: High vibration is associated with deteriorated process performance and energy efficiency — it does not simply reflect poor sensor health (since sensor health shows no correlation with vibration).

8. Energy Efficiency Analysis

State	Avg. Yield	Avg. Energy Intensity	Pressure	Flow	Vibration
High Yield + Low Energy (best)	93.13 tons	2.35	0.89 bar	583 m³/h	3.72 mm/s
Low Yield + High Energy (worst)	67.56 tons	3.46	Lower	Lower	5.36 mm/s

High production does **not** require high energy intensity — the best state proves both are simultaneously achievable.

Sensor health is similar across both states — not a differentiator.

Takeaway: Efficient operation = favourable pressure + flow, combined with avoiding high-vibration conditions.

9. Anomaly Detection

Anomalies found per variable:

Variable	Anomalous Rows
Reactor Temperature	88
Reactor Pressure	0
Vibration Level	0
Sensor Health Index	63
Energy Intensity	152
Product Yield	40

Total distinct anomalous rows: 338

Anomalous observations show **lower yield** (73.3 vs. 80.2 tons overall) and **higher energy intensity** (3.63 vs. 2.89 overall) — unusual operating conditions correlate with worse production performance and efficiency.

10. Overall Synthesis

Despite monitoring 16 variables, only a **small subset — pressure, feedstock flow, and vibration — shows meaningful association with yield.**

Catalyst type, unit, and shift show only small effects (~0.17, ~0.38, and ~0.59 tons respectively) — real, but modest compared to pressure and flow.

The best unit-catalyst-shift combination (Methanol + Platinum + Morning) is itself associated with slightly better pressure, flow, vibration, and energy intensity — suggesting the *combination's* advantage comes from operating conditions and shift timing rather than catalyst identity alone.

Catalyst age has no meaningful relationship with yield or energy intensity, overall or within any catalyst type.

Important caveat carried throughout: All of the above are *associations from historical/observational data* — they do not establish causation.

What Might Still Be Missing / Worth Adding

Your analysis is genuinely strong — you've correctly separated meaningful drivers from noise, checked within-group variation (not just marginal correlations), and consistently flagged the observational-not-causal caveat. A few things would make it more complete and more defensible if a reviewer/professor pushes on it:

Formal significance testing. Right now the catalyst/unit/shift comparisons are based on means and std alone (e.g., Iron vs. Cobalt differ by 0.0004 tons). Run a t-test (two groups) or one-way ANOVA (three groups) to confirm which differences are statistically real vs. just noise. This directly strengthens claims like "catalyst-level yield differs by only ~0.17 tons."

Multicollinearity check between pressure and flow. Since both are strong positive drivers and likely correlated with each other physically, check `corr(pressure, flow)`. If it's high, your r-values for each individually may be overstated, and a combined/partial correlation or regression would be more honest.

A combined regression model. A multiple linear regression (yield ~ pressure + flow + vibration, at minimum) would let you quantify each factor's *independent* contribution once the others are controlled for, and give you an R^2 for how much of yield's variation these 3–4 factors actually explain together. This ties your whole "only 3–4 factors matter" narrative into one clean, quantified statement.

Sample sizes (counts) alongside every mean/std table. You have this for the pressure bins but not consistently for unit/catalyst/shift comparisons — needed before any significance testing, and useful to flag if any group is much smaller than others.

Anomaly detection method. State explicitly which method you used per variable (IQR, z-score, or a fixed threshold) — right now it's implied but not documented, and a reviewer will ask.

The heatmap you're about to build — good addition, this will visually confirm the pressure/flow/vibration-vs-others contrast you've already found numerically.

Quantified cost/business angle. You've discussed a "maximize economically useful yield at minimum cost" recommendation conceptually — if feasible, even a rough estimate (e.g., cost implied by the ~0.59 ton shift effect, or energy savings from the low-vibration regime) would make the business-recommendation section more concrete.

None of these are corrections to what you've done — they're additions that would move the project from "well-executed EDA + correlation analysis" to "EDA + statistically validated conclusions," which is a meaningfully stronger final report.